STAT5080 – HW 5 Solutions

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Problem 1

Four Contrasts of Interest

The four contrasts are defined as follows (using groups 1=Control, 2=Drug A Low, 3=Drug A High, 4=Drug B Low, 5=Drug B High):

Contrast	Description	Coefficients (c1, c2, c3, c4, c5)
C1	Drug A vs Drug B	0, 1, 1, -1, -1
C2	Control vs Combined Experimental	4, -1, -1, -1, -1
C3	Low vs High dosage – Drug A	0, 1, -1, 0, 0
C4	Low vs High dosage – Drug B	0, 0, 0, 1, -1

```
## Contrast estimates, SEs, and t-statistics:
##   c1: estimate = -17.6000, SE = 3.3638, t = -5.2322
##   c2: estimate = -6.4000, SE = 5.8263, t = -1.0985
##   c3: estimate = 6.0000, SE = 2.3786, t = 2.5225
##   c4: estimate = -5.2000, SE = 2.3786, t = -2.1862
```

Part (a): Holm's Step-Down Procedure (= 0.05)

Holm's procedure ranks |t| values from largest to smallest, then compares each to a sequentially adjusted critical value.

Steps:

1. Rank contrasts by |t| from largest to smallest.
2. At step k (starting from k=1), compare the largest remaining |t| to $t_{df_E,\,\alpha/(2(m-k+1))}$.
3. If it exceeds the critical value, reject H and continue; otherwise stop.

```
## Ranked |t| values (descending):
##   Rank 1: c1, |t| = 5.2322
##   Rank 2: c3, |t| = 2.5225
##   Rank 3: c4, |t| = 2.1862
##   Rank 4: c2, |t| = 1.0985
##
## --- Holm's Step-Down Procedure ---
## df(Error) = 25
##
## Step 1: contrast = c1, |t| = 5.2322, critical value = t(25, 0.0125) = 2.6916  -->  REJECT H0
## Step 2: contrast = c3, |t| = 2.5225, critical value = t(25, 0.0167) = 2.5660  -->  Fail to Reject H0 --
```

Summary – Holm's Procedure:

Step	Contrast	$\|t\|$ value	Adjusted α	Critical Value	Decision
1	c1	5.2322	$0.05/4 = 0.0125$	2.6916	REJECTH_0
2	c3	2.5225	$0.05/3 = 0.0167$	2.5660	Fail to RejectH_0 (STOP)
3	c4	2.1862	$0.05/2 = 0.0250$	2.3846	Fail to Reject H_0
4	c2	1.0985	$0.05/1 = 0.0500$	2.0595	Fail to Reject H_0

Part (b): Comparison with Bonferroni's Procedure

Under Bonferroni, all m comparisons use the same critical value $t_{df_E,\,\alpha/(2m)}$ regardless of ranking.

```
## Bonferroni critical value: t(25, 0.0125) = 2.6916
##
## --- Bonferroni Procedure ---
##   c1: |t| = 5.2322  -->  Reject H0
##   c2: |t| = 1.0985  -->  Fail to Reject H0
##   c3: |t| = 2.5225  -->  Fail to Reject H0
##   c4: |t| = 2.1862  -->  Fail to Reject H0
```

Bonferroni vs Holm's Step-Down Comparison

Contrast	$\|t\|$	Bonferroni CV	Bonferroni Decision	Holm Decision
c1	5.2322	2.6916	Reject H_0	Reject H_0
c2	1.0985	2.6916	Fail to Reject H_0	Fail to Reject H_0
c3	2.5225	2.6916	Fail to Reject H_0	Fail to Reject H_0
c4	2.1862	2.6916	Fail to Reject H_0	Fail to Reject H_0

Holm's procedure uses a less conservative (smaller) critical value for contrasts with larger test statistics, making it *more powerful* than Bonferroni. Holm's procedure controls the family-wise error rate at most , just as Bonferroni does, but it can reject more null hypotheses. If the two methods differ, it is because Holm's procedure detected significance for a contrast that Bonferroni did not, a direct consequence of using sequentially adjusted (smaller) critical values for the most extreme test statistics.

Problem 2

Experiment: $a = 4$ groups, $n = 11$ per group $\rightarrow N = 44$, $df_E = N - a = 40$.

The test statistic for the contrast is given as $F = 7.0$.

Part (a): Pre-Planned Contrast

When a single contrast is **pre-planned**, only $m = 1$ comparison is being made. The appropriate critical value is the standard F critical value with 1 numerator degree of freedom and df_E denominator degrees of freedom at level α:

$$F_{1,\,df_E,\,1-\alpha} = F_{1,\,40,\,0.95}$$

```
## Critical value (planned, m=1): F(1, 40, 0.95) = 4.0847
## F = 7.00  >  4.0847
## Decision: SIGNIFICANT (Reject H0)
```

Part (b): Post-Hoc Contrast

When the contrast is identified **after looking at the data**, we must protect against capitalizing on chance. The Scheffé method covers all possible contrasts simultaneously and is appropriate here. The Scheffé critical value (in terms of F) is:

$$(a-1) \cdot F_{a-1,\, df_E,\, 1-\alpha} = 3 \cdot F_{3,40,\, 0.95}$$

```
## Scheffe critical value: 3 * F(3,40,0.95) = 3 * 2.8387 = 8.5162
## F = 7.00  <=  8.5162
## Decision: NOT significant (Fail to reject)
```

Part (c): Explanation of the Discrepancy

The result differs depending on *when* the contrast was chosen because:

- **Pre-planned contrasts** are tested at face value using a standard critical value. Since the researcher committed to this specific comparison *before* seeing the data, there is no inflation of the Type I error, only a single comparison is being made.
- **Post-hoc contrasts** are discovered by scanning the data for interesting patterns. This is a form of *data snooping* : if we examine many possible contrasts and pick the most extreme one to test, we are effectively performing many tests simultaneously, even if w eonly report one. Scheffé's method adjusts for this by requiring the test statistic to exceed a larger critical value, one that controls the family-wise error rate over *all possible contrasts*, not just the one we happened to choose.

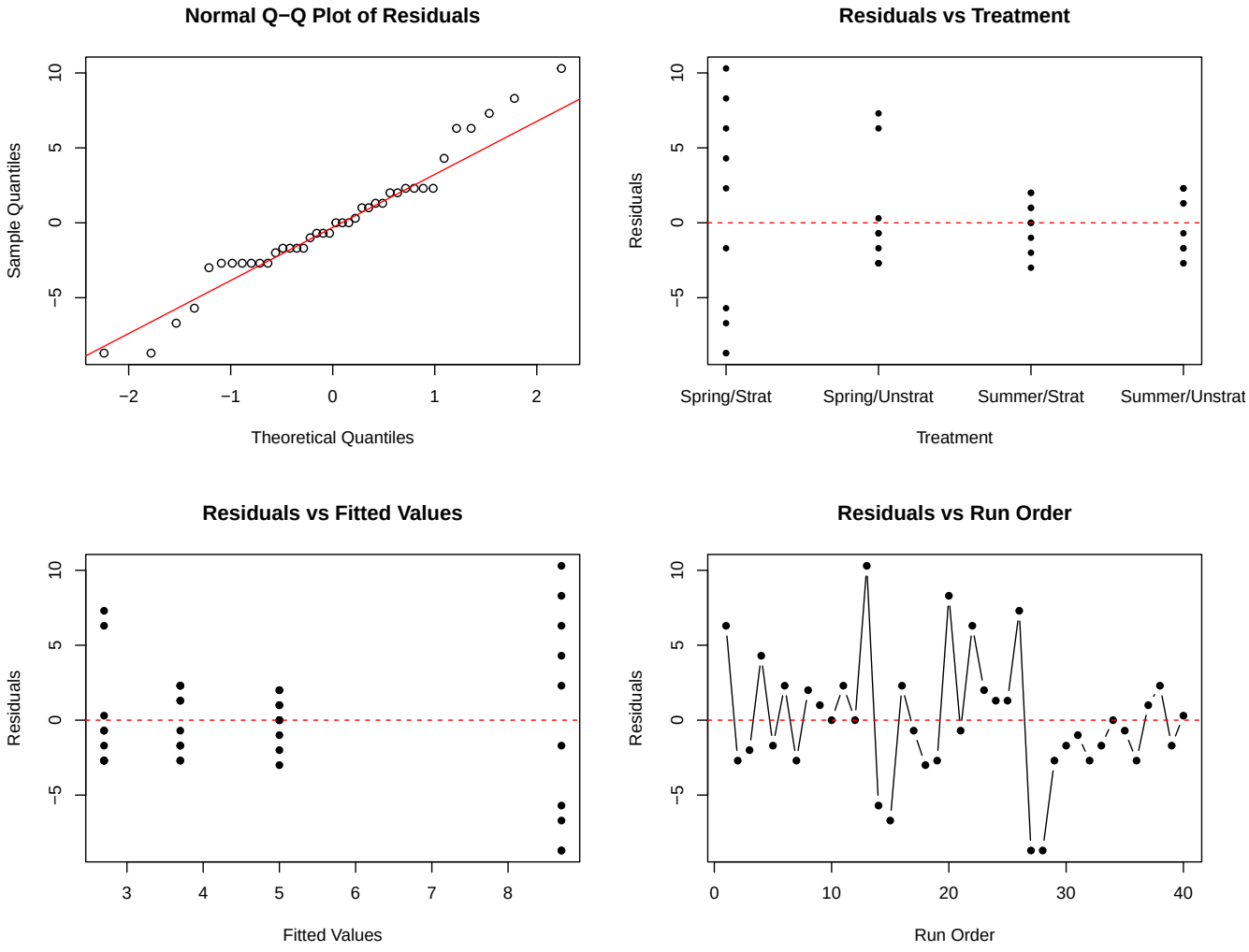
In essence, the significance of a contrast legitimately depends on whether it was pre-specified, because the act of choosing a contrast based on the data inflates the probability of a false positive. The larger Scheffé critical value corrects for this inflation.

Problem 3

Table 4: Summary Statistics of Seed Germination by Treatment

Treatment	Mean	Variance	Median
Spring/Strat	8.7	52.23	9.0
Spring/Unstrat	2.7	14.01	1.5
Summer/Strat	5.0	2.67	5.0
Summer/Unstrat	3.7	4.46	4.0

Part (a): Graphical Residual Analysis



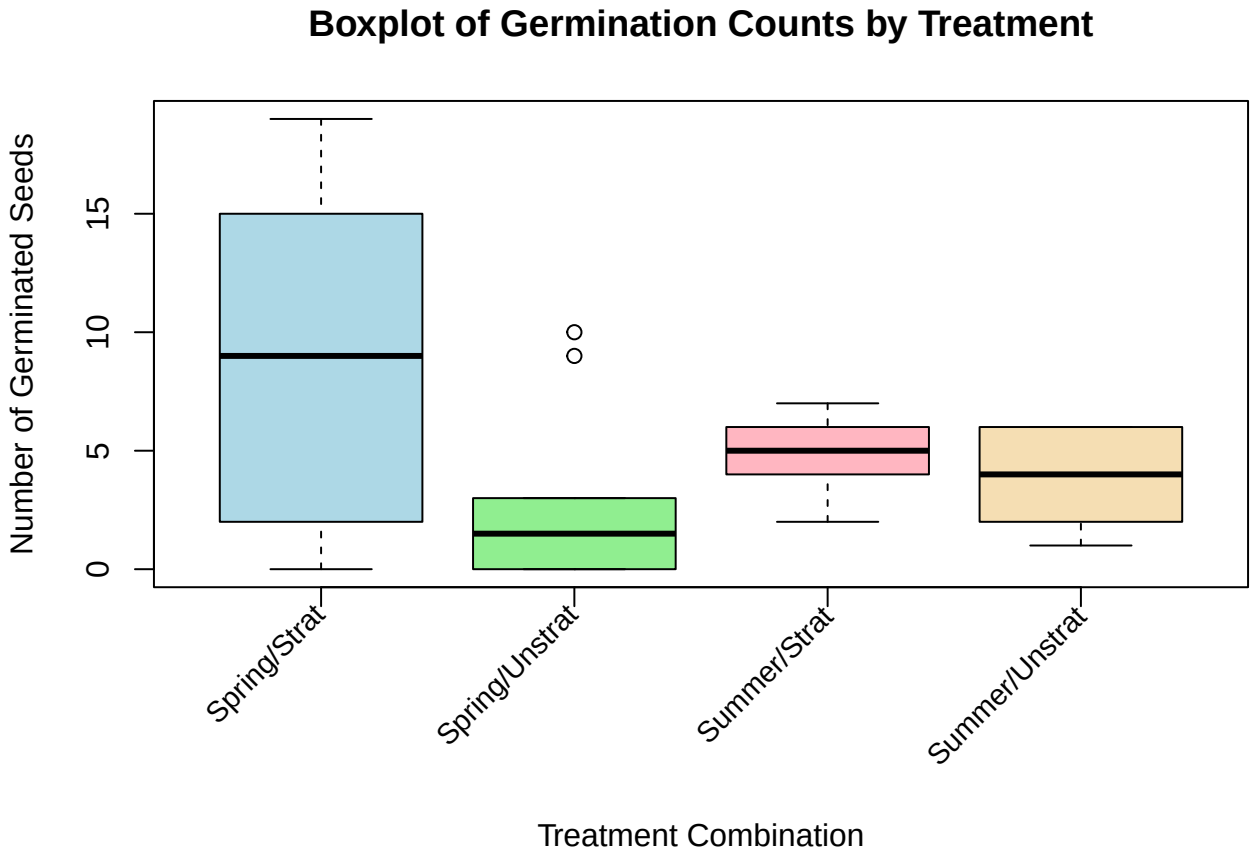
Interpretation:

- **Normal Q-Q plot**: The residuals deviate noticeably from the straight line, especially in the tails, suggesting non-normality. This is expected for count data (number of germinating seeds out of 20).
- **Residuals vs Treatment**: The spread of residuals differs across treatment groups Spring/Stratified shows much higher variance than the other groups indicating *non-constant variance*.
- **Residuals vs Fitted**: A fan-shaped pattern is visible; variance appears to increase with the fitted values, confirming heteroscedasticity.
- **Residuals vs Run Order**: No obvious systematic trend is apparent, suggesting the independence assumption is approximately satisfied.

Conclusion: The normality and constant variance assumptions appear to be violated. The independence assumption looks reasonable.

Part (b): Formal Test of Constant Variance

Graphical Assessment of Variances:



The side-by-side boxplots reveal drastically different spreads across the treatment groups. Specifically, the "Spring/Stratified" group exhibits a massive spread (ranging from 0 to almost 20), whereas the "Spring/Unstratified" and both Summer groups have very tightly clustered data points. This clear visual discrepancy strongly suggests that variance is not constant across treatments.

Levene's Test:

```
## Levene's Test (ANOVA on absolute deviations from group medians):
## Analysis of Variance Table
##
## Response: abs_resid
##           Df Sum Sq Mean Sq F value    Pr(>F)
## trt_germ   3 155.88  51.958  11.227 2.399e-05 ***
## Residuals 36 166.60   4.628
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Bartlett's Test:

```
## Bartlett's Test:
##
##  Bartlett test of homogeneity of variances
##
## data:  y_germ by trt_germ
## Bartlett's K-squared = 22.073, df = 3, p-value = 6.298e-05
```

Because the data consists of discrete counts and deviates from perfect normality, **Levene's Test** (centered on the median) is the most robust and appropriate formal test.

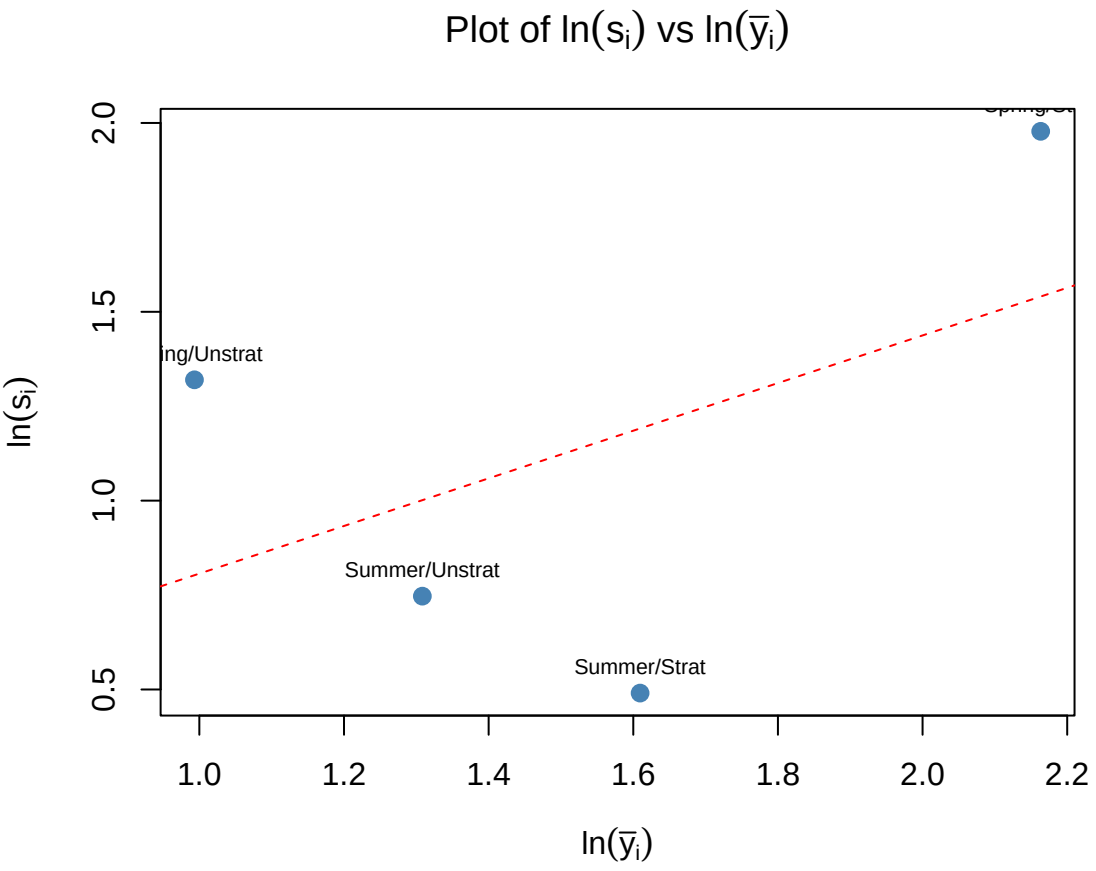
Hypotheses:

- $H_0 : \sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \sigma_4^2$ (All treatment groups have equal variances)
- H_a : At least one treatment group has a different variance.

Results: Levene's test yielded an F-statistic of 11.23 with a highly significant p-value ($p \approx 0.000024$). Bartlett's test corroborates this result ($K^2 = 22.07$, $p \approx 0.000063$). Because the p-value (< 0.001) is significantly less than the standard $\alpha = 0.05$ level, we reject the null hypothesis.

Conclusion: Both the visual evidence from the boxplots and the statistical evidence from the formal tests strongly confirm that the **constant variance assumption is severely violated** in this one-way ANOVA model.

Part (c): Plot of ln(s) vs ln($\bar{y}$.) for Power Transformation



The plot of $\ln(s_i)$ against $\ln(\bar{y}_{i\cdot})$ shows widely scattered points that fail to form a consistent linear trend. Because the variance fluctuates independently of the mean, a general power transformation is "**helpless**" here and will **not equalize** the variances.

Part (d): Kruskal-Wallis Test (Large Sample Approximation)

```
## Ranks:
```

```
##  [1] 38.0  3.5 37.0  3.5 12.5 17.5 32.0 39.0 40.0 36.0 34.0  8.0 12.5  3.5  3.5
## [16] 35.0 12.5  3.5 17.5  3.5 32.0 23.0 20.0 28.0 23.0 23.0 32.0 12.5 28.0 17.5
## [31]  8.0 23.0 28.0 12.5 12.5 17.5 23.0 28.0  8.0 28.0
##
## Rank sums (R_i.):
##   Spring/Strat Spring/Unstrat   Summer/Strat Summer/Unstrat
##          259.0          133.5          239.0          188.5
##
## H (no-ties approximation) = 6.9194
## H (ties-corrected) = 7.0035
##
## Degrees of freedom = 3
## Chi-squared critical value (alpha=0.05) = 7.8147
## p-value = 0.0718
```

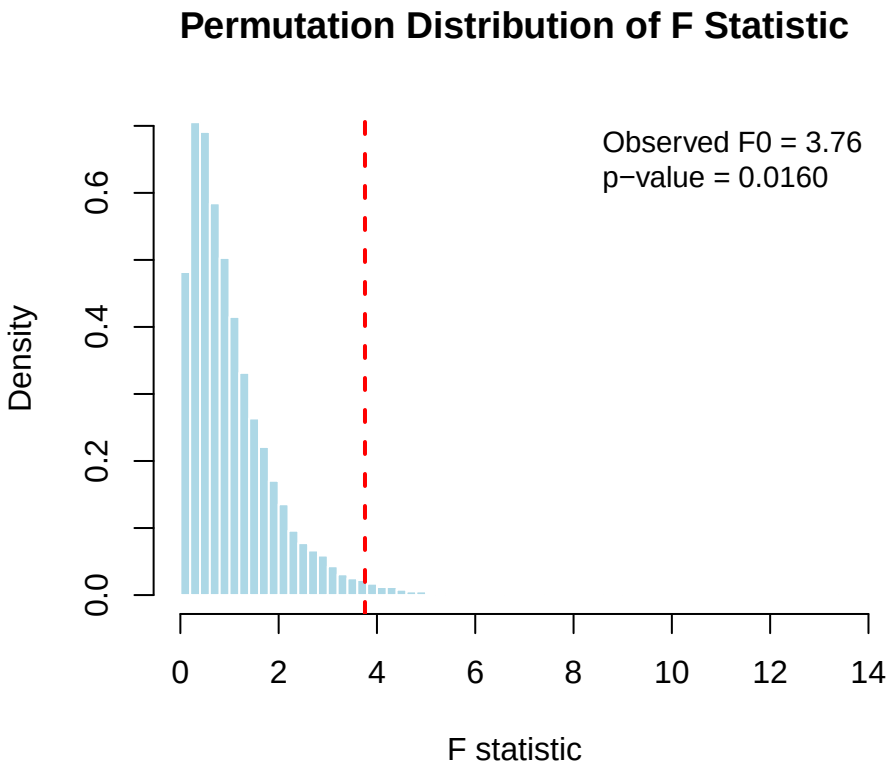
Hypotheses:

- H_0: All four treatment groups have the same distribution (no treatment effect).
- H_a: At least two treatment groups have different distributions.

Conclusion: Since the p-value (0.0718) is greater than the significance level ($\alpha = 0.05$), and the test statistic ($H = 7.0035$) does not exceed the critical value (7.8147), we **fail to reject the null hypothesis**.

There is not sufficient statistical evidence at the 0.05 level to conclude that the temperature and storage treatment combinations significantly affect the germination rate of the seeds.

Part (e): Permutation Test



Hypotheses:

- H_0: No treatment effect (the assignment of observations to groups is arbitrary).
- H_a: At least one treatment group has a different distribution.

Observed Test Statistic: $F_0 = 3.756$

Permutation p-value: 0.0160

Since the permutation p-value (0.0160) is less than the significance level ($\alpha = 0.05$), we **reject the null hypothesis**. We conclude that there is a significant treatment effect on the germination rate. The permutation test is valid regardless of whether the normality or constant-variance assumptions hold, making its conclusion much more reliable for this dataset given the severe violations we observed in Part (a) and (b).